

The six-equation two-phase model as implemented in CAMR

Governing equations, closures and source terms

Generated from CAMR branch `co2-eos`

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Abstract

This note records the two-phase model *as the code actually solves it*, rather than in its textbook form. Each equation, closure and source term is anchored to `file:line`. Runtime options are given with their defaults, because several of the model's decisions are dials rather than constants. Two places where the implementation is known to depart from the model it intends are flagged explicitly in §7.

1 State vector

The conserved state carries the two-phase slots `Utils/IndexDefines.H:145--149` alongside CAMR's ordinary single-fluid slots `:17--18, 74--76`:

$$\mathbf{U} = \left(\underbrace{\rho, \rho\mathbf{u}, \rho E, \rho e, T}_{\text{stock CAMR}}, \underbrace{\alpha_1, \alpha_1\rho_1, \alpha_2\rho_2, \alpha_1\rho_1 E_1, \alpha_2\rho_2 E_2}_{\text{PS extension}} \right)^\top. \quad (1)$$

Write $m_k = \alpha_k \rho_k$ for the partial masses and $\mathcal{E}_k = \alpha_k \rho_k E_k$ for the phase total energies, with $E_k = e_k + \frac{1}{2}|\mathbf{u}|^2$. The volume fractions close as $\alpha_1 + \alpha_2 = 1$, so only α_1 is stored.

The state is deliberately redundant. Mixture mass and energy are stored *both* as ρ , ρE *and* as the phase sums. The two identities

$$\rho = m_1 + m_2, \quad \rho E = \mathcal{E}_1 + \mathcal{E}_2 \quad (2)$$

are therefore invariants to be *maintained*, not consequences of the discretisation. They are re-established every step by `ps_resync_mixture_mass` and `ps_resync_phase_energy` `CAMR_advance.cpp:410--413` and monitored as checks V4 and V5 of `ps_validate_state`. Historically the mass identity drifted at $\sim 0.4\%$ per production run before anyone was watching it.

2 Governing equations

With a single mixture velocity $\mathbf{u}$ (velocity equilibrium) and instantaneous mechanical relaxation, the system solved is

$$\frac{\partial \alpha_1}{\partial t} + \mathbf{u} \cdot \nabla \alpha_1 = \mathcal{S}_\alpha, \quad (3)$$

$$\frac{\partial m_1}{\partial t} + \nabla \cdot (m_1 \mathbf{u}) = \Gamma, \quad (4)$$

$$\frac{\partial m_2}{\partial t} + \nabla \cdot (m_2 \mathbf{u}) = -\Gamma, \quad (5)$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I}) = \mathbf{0}, \quad (6)$$

$$\frac{\partial \mathcal{E}_1}{\partial t} + \nabla \cdot (\mathcal{E}_1 \mathbf{u} + \alpha_1 P_1 \mathbf{u}) = -P_I \frac{\partial \alpha_1}{\partial t} + \mathcal{Q} + \Gamma H_I, \quad (7)$$

$$\frac{\partial \mathcal{E}_2}{\partial t} + \nabla \cdot (\mathcal{E}_2 \mathbf{u} + \alpha_2 P_2 \mathbf{u}) = +P_I \frac{\partial \alpha_1}{\partial t} - \mathcal{Q} - \Gamma H_I, \quad (8)$$

together with the mixture total energy in strict conservation form,

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + P) \mathbf{u}] = 0, \quad (9)$$

which is the quantity that is fluxed and refluxed, with (2) tying it to (7)–(8). The mixture pressure is the volume-fraction average

$$P = \alpha_1 P_1 + \alpha_2 P_2, \quad \text{PS_hllc.H:289} \quad (10)$$

and the interfacial pressure is the arithmetic mean

$$P_I = \frac{1}{2} (P_1 + P_2). \quad \text{hem_pelanti_shyue.H:2157} \quad (11)$$

Equations (3), (7) and (8) are *not* in divergence form: α_1 is advected rather than fluxed, and the phase energies carry the non-conservative product $P_I \partial_t \alpha_1$. That is the entire reason the code uses wave propagation rather than flux differencing for those three slots (§6).

3 Equation of state and the branch-locked contract

The reference backend is Peng–Robinson for pure CO₂ ($T_c = 304.13$ K, $P_c = 7.3773 \times 10^6$ Pa, $\omega = 0.22394$). Queries are *branch-locked*: the caller nominates the phase, and the EOS returns that branch, continued metastably past its saturation limit if necessary. It never auto-detects, because a nominal vapour sitting at compressed-liquid density would otherwise return the liquid root and drive the relaxation toward an unphysical equilibrium.

The corresponding anti-requirement is load-bearing:

the hydraulic path must not ask the EOS about a mixture state.

For a two-phase cell the pair (ρ, e) combines the light phase’s volume with the heavy phase’s mass and need not be a single-fluid state at all, so the inversion may have no root. Temperature inversion brackets $T \in [1, 5000]$ K and *refuses* outside it rather than returning the nearest reachable state.

4 Source terms

The default relaxation is the **coupled X3 operator** (`ps_relax_mode=5`, the default and the acceptance configuration since 2026-08-17; an unrecognised mode *aborts* rather than falling

through `PS_relaxation.H:1030--1050`). Per reaction substep `CAMR_advance.cpp:640--665`:

$$\text{hydro} \longrightarrow \underbrace{\text{vanish fold}}_{\text{phase death}} \longrightarrow \underbrace{\text{X3}}_{\substack{P_1=P_2 \text{ DAE} + \text{thermal} + \\ \text{BE-SRT mass transfer}}} \longrightarrow \underbrace{\text{flash}}_{\text{nucleation}} \longrightarrow \underbrace{\text{mech. reproject}}_{\text{ps_src_p_reproject}=1} \quad (12)$$

X3 solves the cell as a constrained system: mechanical equilibrium $P_1 = P_2$ is imposed as a DAE constraint while the thermal and mass-transfer legs run as backward-Euler rates *on that constrained manifold* — one coupled solve, not a sequence, so the operators cannot fight over α_1 . Its 0-D fixed point is the full-equilibrium (HEM) flash. A closing mechanical pass (`ps_mech_close=1` default) re-establishes $P_1 = P_2$ at chain exit. The historical sequential chain (mechanical $\rightarrow$ thermal $\rightarrow$ flash $\rightarrow$ MT with a re-projection after each of the last two) survives as `ps_relax_mode=4`; mode 0 is mechanical-only and must now be asked for explicitly. In mode 5 the split mass-transfer source is *owned by X3*: enabling it separately aborts `CAMR_advance.cpp:139--147`.

4.1 Mechanical relaxation: $\mathcal{S}_\alpha$

Pressure equilibrium is **instantaneous by model closure**: α_1 is the free variable and is driven until $P_1 = P_2$. It is a constraint, not a rate. The accompanying interfacial work is applied to the phase energies `hem_pelanti_shyue.H:2223--2225`:

$$e_1 \mapsto e_1 - \frac{P_l \Delta \alpha_1}{\alpha_1 \rho_1}, \quad e_2 \mapsto e_2 + \frac{P_l \Delta \alpha_1}{\alpha_2 \rho_2}, \quad (13)$$

equal and opposite in $\mathcal{E}_k$, so mixture energy is preserved exactly. This is the discrete realisation of $\mp P_l \partial_t \alpha_1$ in (7)–(8). The Newton step on α_1 is capped at a fraction of $\min(\alpha_1, 1 - \alpha_1)$ and restricted to the interval where both ρ_k remain inside the EOS domain.

4.2 Thermal relaxation: $\mathcal{Q}$

Finite-rate, isochoric, at the α_1 the mechanical pass chose. It redistributes the energy split only, driving $T_1 \rightarrow T_2$ at rate $1/\theta$ `hem_pelanti_shyue.H, ps_iso_thermal_relax_cell_finite`:

$$\mathcal{E}_k^{n+1} = \mathcal{E}_k^* + (\mathcal{E}_k^{\text{eq}} - \mathcal{E}_k^*) (1 - e^{-\Delta t / \theta}), \quad (14)$$

where $\mathcal{E}^{\text{eq}}$ is the isochoric $T_1 = T_2$ state. Note the form: this is an *exponential relaxation onto a precomputed equilibrium*. `CAMR.ps_theta_tau defaults to  $10^{-7}$  s` (THETA-DEFAULTS 2026-08-24) — near-instantaneous at hydrodynamic scales — and its semantics are: $\theta > 0$ a finite backward-Euler rate; $\theta \leq 0$ *instantaneous* thermal equilibrium, i.e. in mode 5 the constraint $T_1 = T_2$ joins the $P_1 = P_2$ DAE (“ $\theta \leq 0$ does not mean off” `PS_relaxation.H:1139--1151`).

The leg is gated by a two-phase coexistence test — both phase temperatures strictly inside $(T_{\text{triple}}, T_{\text{crit}}) = (216.592, 304.13)$ K — shared with mass transfer and the flash. `CAMR.ps_coexist_action` (default 0) selects what happens when a phase leaves that band: 0 silent no-op, 1 abort, 2 run the thermal leg for low-side exits, 3 run it for any exit. Since 2026-08-27 the gate carries a **runaway clause** `hem_pelanti_shyue.H:2763, PS_relaxation.H`: a *high-side-only* band exit with a phase beyond $2T_{\text{crit}}$ runs the thermal leg regardless, because the alternative — vetoing thermal while the mechanical projection keeps pumping interfacial work into a vanishing sliver — was measured to ratchet e_2 to the EOS band edge at the jet contact (WP-CONTACT-CEIL, WORKLOG 2026-08-27). Runaway activations are counted (`n_exit_runaway`, exit-code bit 8) and printed in `[PS-COEXIT-TH]`.

(The per-cell HRM-correlation option `CAMR.ps_theta_model=1` — Downar-Zapolski $\theta = \theta_0 \alpha_v^{-0.54} \psi^{-1.76}$ — was **retired** 2026-08-17 (S3): measured to be the wrong mapping for *thermal* contact between phases, since the correlation slows toward saturation, which is right for phase change and wrong for heat exchange; B9 aborts under it. The correlation itself remains relevant to mass transfer; see §7(ii).)

4.3 Mass transfer: Γ and H_I

Driven by the Gibbs free-energy difference $g_1 - g_2$ with $g_k = h_k - T_k s_k$, and integrated as an exact relaxation onto the instantaneous mass-transfer equilibrium:

$$\delta m = \left(1 - e^{-\Delta t/\tau}\right) \delta m_{\text{eq}}, \quad \delta m_{\text{eq}} : g_1 = g_2, \quad (15)$$

so $\Gamma = -\delta m/\Delta t$ in (4). The target δm_{eq} is found by a Newton iteration on δm . The form (15) is unconditionally stable and cannot overshoot δm_{eq} — *provided* the target is the equilibrium of the path actually integrated; see §7.

The default rate is derived, not tuned (BE-SRT). At the default (mode 5, `ps_mt_tau_model=2`, ASY1) no relaxation time is chosen: τ in (15) is *computed* as the distance to equilibrium over the instantaneous physical rate, per Lund & Aursand,

$$\tau_{\text{asy}} = \frac{|\delta m_{\text{eq}}|}{\Gamma_{\text{SRT}}}, \quad \Gamma_{\text{SRT}} = \Sigma \rho_g \sqrt{\frac{M}{2\pi R_u T_g}} |g_1 - g_2|, \quad \Sigma = \frac{16}{\pi} \frac{(\alpha_{\text{recv}} + \delta) \alpha_{\text{send}}}{D}, \quad (16)$$

`hem_pelanti_shyue.H:2364--2407`: the kinetic prefactor is parameter-free statistical rate theory; the *interfacial-area* closure Σ is not — it carries the stratified-pipe morphology placeholder $D = \text{CAMR.ps_mt_srt_d}$ (default 0.1 m) and the receiving-phase seed $\delta = \text{CAMR.ps_mt_srt_delta}$ (default 0.01). A degenerate Γ falls back to the constant- τ form so the dial can never invent a rate. The morphology dependence this hides is quantified in §7(ii). The constant `ps_mt_tau` route belongs to the non-default modes.

Transferred mass carries the interface total enthalpy

$$H_I = h_I + \frac{1}{2} |\mathbf{u}|^2, \quad h_I = \begin{cases} (1-w) h_1 + w h_2, & w \geq 0 \quad (w = \frac{1}{2} \text{ default: mean}), \\ h_1 \text{ if } \delta m > 0, \quad h_2 \text{ otherwise,} & w < 0 \quad (\text{upwind}), \end{cases} \quad (17)$$

selected by `CAMR.ps\_mt\_h\_weight`.

Volume co-moves with the mass (E.1). Under the presence model the transfer moves volume at the *donor's own* density:

$$\Delta \alpha_1 = - \frac{\delta m}{\rho_1}. \quad (18)$$

Two consequences make this the derived choice rather than a convenience. The donor's density is then *exactly* invariant, $\rho'_1 = (m_1 - \delta m)/(\alpha_1 - \delta m/\rho_1) = \rho_1$, so it can never leave the EOS domain from below; and the receiver's density becomes a convex combination (a median) of ρ_2 and ρ_1 , so it can never leave from above. The proof is rate-independent: even $\delta m = m_1$ in one step lands α_1 exactly on zero.

4.4 Flash: nucleation only

The flash is **not** a conversion operator. It seeds a minority phase up to $\alpha_{\text{birth}} = 2 \times 10^{-2}$ and declines once that level is reached `hem_pelanti_shyue.H:3778`:

$$\alpha_2 \mapsto \alpha_2 + \beta (\alpha_{\text{birth}} - \alpha_2), \quad \text{return if } \alpha_2^{\text{new}} \leq \alpha_2, \quad (19)$$

with β a blend of a metastability window, a dominance ramp and $\Delta t/\tau_f$. Newborn mass is injected at the *saturation* density ρ_{sat} , which is (18) run backwards, so birth and death are the same operator with opposite sign. Past α_{birth} , conversion is mass transfer's job alone.

5 Presence: when a phase has a state at all

The intensive quantities are quotients with *different* denominators,

$$\rho_k = \frac{m_k}{\alpha_k}, \quad e_k = \frac{\mathcal{E}_k}{m_k} - \frac{1}{2}|\mathbf{u}|^2, \quad (20)$$

so a test on one does not license the other. Because the hydro advances m_k with an absolute error set by the *mixture* scale, ρ_k inherits a relative error $\sim \eta/\alpha_k$. A phase is therefore assigned one of three regimes by α_k alone — no auxiliary flag array, so the regime cannot fall out of step with the state:

regime	α_k	meaning
ABSENT	exactly 0	no intensive quantity is ever formed; the cell is single-phase Euler
CORRIDOR	$(\alpha_{\text{van}}, \alpha_{\text{cond}})$	transport-only: advects, but no EOS query, no relaxation, no transfer
INDEPENDENT	$[\alpha_{\text{cond}}, 1 - \alpha_{\text{cond}}]$	a full six-equation phase; every operator acts

with $\alpha_{\text{cond}} = 2 \times 10^{-2}$ (a *conditioning* bound, re-derived 2026-08-15 as $\eta_{\text{max}}/\varepsilon$ from a measured per-step $\eta_{\text{max}} = 5.2 \times 10^{-4}$ at a 5% accuracy target), $\alpha_{\text{van}} = 10^{-8}$ (death), and $\alpha_{\text{birth}} = 2\alpha_{\text{cond}}$ (nucleation seed). The ordering gives hysteresis with no flip-flop channel: birth at $2\alpha_{\text{cond}}$, loss of independence only below α_{cond} , death only below α_{van} .

6 Discretisation of the non-conservative slots

Since 2026-08-27 the interior scheme is **single-path**: wave propagation (`CAMR.ps_flux=wp`, the Berger–LeVeque fluctuation form) is the only interior flux. `llf`, `hllc`, a set `ps_ctu`, `ps_recon` or `ps_alpha_limiter` key all *abort* with retirement notes `PS_umeth.cpp:5--12,727--745`; `PS_reconstruction.H` is deleted — the wp interior works from raw cell averages. Under AMR the fluctuation form re-fluxes through the standard register with α moved alongside (`CAMR.ps_bl_reflux=2`, the default since 2026-08-26).

One Riemann solve per face produces a wave decomposition (retained in `PS_hllc.H`, which now serves the wp fluctuations), from which the code takes *two different exits*.

Conserved slots $(\rho, \rho\mathbf{u}, \rho E, m_1, m_2)$ receive a single recovered interface flux $F^* = F(\mathbf{U}_L) + \mathbf{A}^-\Delta Q$, which telescopes exactly to the fluctuation update, so CAMR’s ordinary flux-difference path and its AMReX flux register are used unchanged.

Non-conserved slots $(\alpha_1, \mathcal{E}_1, \mathcal{E}_2)$ get a face flux of *exactly zero* and a direct fluctuation deposit,

$$\left. \frac{d\alpha_1}{dt} \right|_i = - \frac{\mathbf{A}^+\Delta Q_{i-1/2} + \mathbf{A}^-\Delta Q_{i+1/2}}{\Delta x}, \quad (21)$$

and likewise for $\mathcal{E}_1, \mathcal{E}_2$. Because both exits are fed by the same wave structure they cannot disagree about what happened at the face, which is what makes a uniform-pressure, uniform-velocity interface drift without generating spurious waves *by construction* rather than by cancellation between two independent discretisations.

6.1 Wave structure

Three waves per face (HLLC, specialised to the six-equation system following Pelanti 2022). Outer speeds are Davis estimates using each side’s own mixture sound speed `PS_hllc.H:375--376`:

$$S_L = \min(u_L - c_L, u_R - c_R), \quad S_R = \max(u_L + c_L, u_R + c_R), \quad (22)$$

with the Wallis frozen mixture speed `PS_hllc.H:303`

$$c^2 = Y_1 c_1^2 + Y_2 c_2^2, \quad Y_1 = \frac{\alpha_1 \rho_1}{\rho}, \quad Y_2 = 1 - Y_1. \quad (23)$$

“Frozen” means no mass transfer during the acoustic response; the much slower equilibrium speed belongs to the relaxation operator, which is applied separately. (`CAMR.ps_cmix_model`, which offered $c = \max(c_1, c_2)$ and the Wood equilibrium form, was measured 2026-08-13 and **retired** 2026-08-17: MAX gains 3% on a contact because S_L, S_R only bound the fan, and WOOD aborts B9 with a real-fluid EOS `PS_presence.H:127--136`. The frozen mass-weighted form is now the only form.)

The contact speed follows from the mixture mass and momentum balance `PS_hllc.H:386`:

$$S_M = \frac{P_R - P_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)}. \quad (24)$$

α_1 is advected at S_M — the local contact velocity at that face, not a cell-averaged mixture velocity — which is what preserves the contact.

6.2 Star states

A *single* contraction ratio, shared by both phases `PS_hllc.H:491--495`:

$$r_K = \frac{S_K - u_K}{S_K - S_M}, \quad m_1^* = r_K m_1, \quad m_2^* = r_K m_2. \quad (25)$$

Because both partial masses scale by the same number, their ratio — and hence the composition — is untouched by the acoustic compression, which is what lets the volume fraction survive the fan. The per-phase star energies are `PS_hllc.H:515--519`

$$E_k^* = E_k + (S_M - u_K) \left(S_M + \frac{P_{e,k}}{q_k} \right), \quad q_k = \rho_k (S_K - u_K), \quad (26)$$

with $P_{e,k} = P$, the *mixture* pressure. (The per-phase alternative `CAMR.ps_pk_energy_flux=1` was **retired** 2026-08-26; a set key aborts `PS_umeth.cpp:843--855`. The two coincide at mechanical equilibrium anyway.) The asymmetric pairing of a mixture pressure with a per-phase density in (26) is COTT’s `HLLC.v` form — a validated choice, not an obvious one.

Failure is refusal, not repair. A non-finite intermediate, a vanishing denominator or a non-positive star density makes the face return `false`, and that face alone falls back to a local Lax–Friedrichs flux. Nothing is clamped to make the star state exist.

7 Two known departures of the implementation from the model

Recorded here because they are properties of the code, not of the model. Both were identified on 13 August 2026; their status as of 28 August:

(i) **The mass-transfer target and the step’s path (resolved by default).** The step applies (18) with the carrier (17); historically δm_{eq} in (15) was computed at *fixed* α_1 with a *mean* carrier regardless of w . Along fixed α_1 , $\partial \rho_1 / \partial m = 1/\alpha_1$, whereas along (18) it is exactly zero, so the target’s Gibbs sensitivity carried a $1/\alpha_1$ amplification the step does not have, breaking the non-overshoot property of (15). `CAMR.ps_mt_target now defaults to 1` (consistent target: same carrier and the same (E.1) path as the step `PS_sources.H:355--362`); 0 retains the legacy behaviour for A/B.

(ii) **The mass-transfer rate carries a morphology assumption — now with a measured handle.** The physical rate is set by the interfacial area the two phases share *within* a cell; the six-equation state carries no interfacial-area variable, so the SRT closure (16) parameterises it through the single length D . A dispersed mixture and a numerically smeared contact can carry identical cell states and still require rates differing by orders of magnitude; the coexistence gate of §4 acts as a crude binary proxy for that distinction. Two measurements now calibrate the dial (WORKLOG 2026-08-27/28, the DT7-4.1 shock tube of SINTEF report DT7 2019-06): (a) the *effective* HRM relaxation time of the scheme maps linearly onto the report’s θ -axis as $\theta_{\text{eff}} \approx 0.08 D$ [s per m], verified over four decades including the slow-branch topology change, so the default $D = 0.1$ m behaves as $\theta_{\text{eff}} \approx 8$ ms; and (b) the Downar-Zapolski correlation evaluated at that flow’s own fan states prescribes $\theta \sim 2 \times 10^{-5}$ s — the default morphology placeholder is ~ 2.5 decades *slower* than the flashing-flow correlation on that problem. A DZ-defensible default would be $D \sim 2.5 \times 10^{-4}$ m; changing it is an open (G-DEF-gated) decision, not yet taken.

8 Characteristic outflow (NSCBC) with a flashing ghost

Subsonic characteristic outflow (`CAMR.ps_bc_use_nscbc=1`; construction in `PS_nscbc.H`) builds the ghost state from the outgoing invariant R^+ (minmod-extrapolated) and an ambient-encoding target $R_{\text{targ}}^- = -P_{\text{amb}}/\rho c$, with a restoring pull $\sigma = 0.25$; the ambient pressure is `prob.p_amb`, overridable *per face* via `CAMR.ps_bc_p_amb_{x,y,z}{lo,hi}` (sentinel -1 inherits). The per-phase repack integrates the isentrope adaptively in pressure sub-steps capped at $\Delta P \leq 0.05 \rho_k c_k^2$ (one step reproduces the historical small-signal arithmetic bit-for-bit). For large exterior jumps the ghost is a **choked fan**: the state marches toward P_{amb} along the per-phase isentropes with HEM saturated re-closure inside the dome (T_{sat} by bisection on the EOS P_{sat} ; enthalpy-lever quality), $du = -dP/(\rho c_{\text{path}})$ with $c_{\text{path}}^2 = dP/d\rho_{\text{mix}}$ along the path, terminating at the critical-flow throat ($G = \rho u$ maximum) or at P_{amb} , whichever comes first. `CAMR.ps_bc_nscbc_flash=1` (default) enables the in-dome re-closure; `0` reproduces the frozen construction for A/B; other values abort. Measured on the flashing-vent benchmark: vented-mass deficit -4.1% against a boundary-free reference (the retired legacy construction sat at -7.6% , the frozen one at -34%). The legacy treatment and its `ps_bc_nscbc_v2` key are retired (a set key aborts).

Summary of runtime options referenced above

parameter	default	selects
<code>ps_flux</code>	<code>wp</code>	the interior scheme (only value; others abort)
<code>ps_bl_reflux</code>	2	AMR reflux handling of the wp fluctuations
<code>ps_relax_mode</code>	5	X3 coupled relaxation (12); 4 = legacy chain
<code>ps_theta_tau</code>	10^{-7} s	θ in (14); ≤ 0 = joint T constraint
<code>ps_mt_srt_d</code>	0.1 m	morphology length D in (16)
<code>ps_mt_srt_delta</code>	0.01	receiving-phase seed δ in (16)
<code>ps_mt_h_weight</code>	0.5	the carrier w in (17)
<code>ps_mt_update_alpha</code>	on	the (E.1) rule (18)
<code>ps_mt_target</code>	1	consistent target, §7(i); 0 = legacy
<code>ps_coexist_action</code>	0	response to a band exit (runaway clause always active)
<code>ps_mech_kernel</code>	0	mode-0 mechanical kernel; 1 = COTT impedance-weighted
<code>ps_mech_close</code>	1	closing mechanical pass at chain exit
<code>ps_src_p_reproject</code>	1	re-project $P_1 = P_2$ after flash/MT
<code>ps_flash_tau</code>	0 (off)	enables the flash nucleator
<code>ps_alpha_cond</code>	2×10^{-2}	the conditioning bound (re-derived 2026-08-15)
<code>ps_presence_vanish</code>	10^{-8}	the death edge (vanish fold threshold)
<code>ps_bc_use_nscbc</code>	0	characteristic outflow, §8
<code>ps_bc_nscbc_flash</code>	1	HEM re-closure along the boundary fan

Retired keys (a set key aborts with a retirement note). `ps_recon`, `ps_ctu`, `ps_alpha_limiter`, `ps_flux=llf/hllc`, `ps_pk.energy_flux`, `ps_cmix_model`, `ps_theta_model`, `ps_hrm_theta`, `ps_alpha_vanish` (superseded by `ps_presence_vanish`), `ps_bc_nscbc_v2`. Retirement-by-abort rather than silent acceptance is deliberate: a stale deck fails loudly instead of running a configuration that no longer exists.